

Guided Lab 6.1 — README Instructions

Course 02 – Introduction to Machine Learning for Cybersecurity

Module 06: Machine Learning for Cybersecurity



Guided Lab 6.1

Detecting Suspicious Cybersecurity Behavior with Machine Learning



Lab Goal

In this guided lab, students will apply machine learning techniques to identify suspicious or anomalous cybersecurity behavior using a cybersecurity or fraud detection dataset.

Students will:

- Explore cybersecurity-focused data
- Prepare data for machine learning analysis
- Train machine learning models
- Detect suspicious or anomalous behavior
- Evaluate model performance
- Interpret findings in a cybersecurity context

This lab connects machine learning concepts directly to cybersecurity monitoring and threat detection applications.

Note

The goal of this lab is not only to build a machine learning model, but also to understand:

- How AI supports cybersecurity systems

- How suspicious behavior can be identified through data
- How machine learning models are evaluated
- How AI-driven systems support threat detection and monitoring

Dataset

Use the provided dataset:

 [creditcard.csv](#)

Dataset Description

- `Class = 0` → Normal transaction
- `Class = 1` → Fraudulent transaction

This dataset simulates fraud detection and anomaly detection scenarios commonly used in cybersecurity analytics.

✂ Lab Activities

Part 1 — Import Libraries

Students will:

- Import Python libraries
- Import machine learning libraries
- Import visualization tools

Example Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Part 2 — Load and Explore the Dataset

Students will:

- Load the dataset into Python
- Review dataset structure
- Check for missing values
- Examine class distribution
- Explore suspicious activity patterns

Part 3 — Prepare the Data

Students will:

- Define features (x) and target (y)
- Split the dataset into training and testing sets
- Scale or preprocess data if necessary

Part 4 — Train a Machine Learning Model

Students will:

- Train at least one machine learning model

Suggested Models:

- Logistic Regression
- Random Forest
- Decision Tree
- Isolation Forest (optional)

Part 5 — Detect Suspicious Behavior

Students will:

- Generate predictions
- Identify potentially suspicious or fraudulent transactions
- Compare predicted results to actual results

Part 6 — Evaluate Model Performance

Students will:

- Calculate:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
- Create a confusion matrix
- Interpret evaluation results

Part 7 — Visualize Findings

Students will:

- Create visualizations
- Explore suspicious behavior patterns
- Interpret trends and anomalies

Possible Visualizations:

- Confusion matrix heatmaps
- Class distribution graphs
- Fraud vs. normal transaction comparisons

Part 8 — Interpretation and Reflection

Students should reflect on:

- Model strengths and weaknesses
- False positives and false negatives
- Cybersecurity implications of suspicious behavior detection
- The role of AI in cybersecurity systems



Reflection Questions

Answer the following questions in complete sentences:

1. How accurate was the machine learning model?
2. What types of suspicious behavior were identified?
3. What do precision and recall tell us about threat detection performance?
4. What does the confusion matrix reveal about the model?
5. Why are false negatives dangerous in cybersecurity systems?
6. How can machine learning improve cybersecurity monitoring?
7. What are some limitations of AI-driven cybersecurity systems?



Submission Requirements

Submit:

- ✓ Completed Jupyter Notebook (.ipynb)
- ✓ All code cells executed with outputs visible
- ✓ Evaluation metrics included
- ✓ Confusion matrix included
- ✓ Visualizations included
- ✓ Reflection questions answered



Important Notes

- Update the dataset path if needed
- Use comments throughout the notebook to explain your workflow
- Carefully interpret suspicious activity predictions
- Focus on understanding evaluation metrics, not only generating outputs



Tips for Success

- Pay close attention to precision and recall
- Think critically about false positives and false negatives
- Carefully analyze suspicious activity predictions
- Use visualizations to support your interpretation
- Connect findings to cybersecurity monitoring and threat detection concepts



Cybersecurity Connection

Machine learning systems are increasingly used in:

- Fraud detection systems
- Intrusion detection systems (IDS)
- Threat intelligence platforms
- Security Operations Centers (SOCs)
- User behavior analytics systems

This lab demonstrates how machine learning can help cybersecurity analysts identify suspicious activity more efficiently.